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EDUCACIÓN PARA AVANZAR

SEMILLERO: PRUEBA DE ANALISIS Y CONFIGURACIÓN.

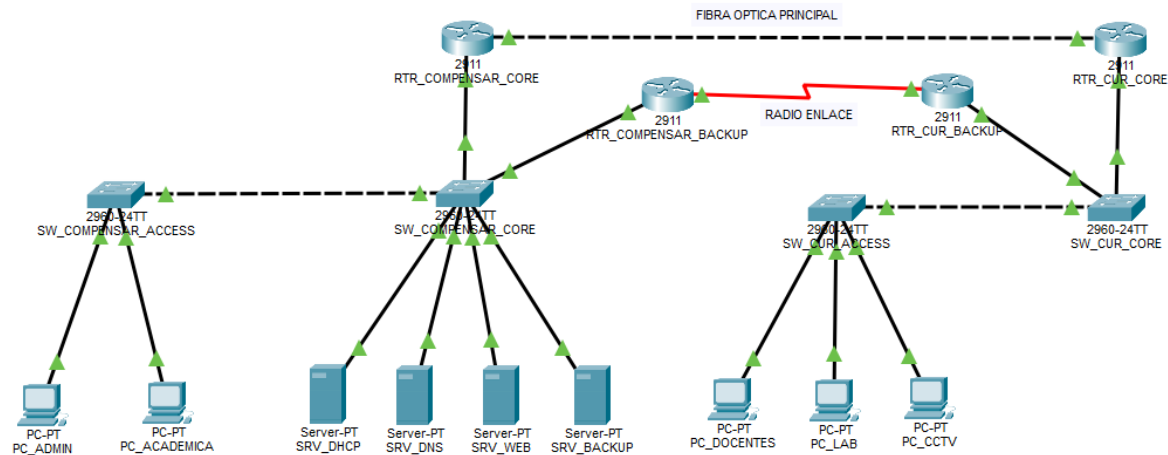
Manual de Configuración, Implementación y Pruebas de la
Arquitectura de Interconexión Institucional

INTEGRANTE

JUAN MELQUICEDET GALINDO JOYA



MONTAJE DE CONEXIÓN ENTRE LA INFRAESTRUCTURA DE LA FUNDACION UNIVERSITARIA COMPENSAR Y LA C.U.R COMPENSAR



La arquitectura híbrida propuesta fue diseñada para mejorar la comunicación entre la Fundación Universitaria Compensar y la C.U.R. de la Universidad Compensar mediante una infraestructura de red más segura, escalable y disponible. La combinación de fibra óptica como enlace principal y un radioenlace como respaldo permite garantizar la continuidad de los servicios institucionales, reducir la dependencia del internet público y optimizar el intercambio de información entre ambas ubicaciones. Adicionalmente, la implementación de VLANs, enrutamiento dinámico, servicios centralizados y mecanismos de seguridad contribuye a mejorar el desempeño, la administración y la confiabilidad de la red institucional.

VALIDACIÓN DE VLANs

SW_COMPENSAR_CORE

```

Switch con0 is now available.

Press RETURN to get started.

Switch>show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                           Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Gig0/1, Gig0/2
10   ADMIN                   active
20   ACADEMICA               active
80   SERVIDORES              active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch>

```

SW_CUR_CORE

```

Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

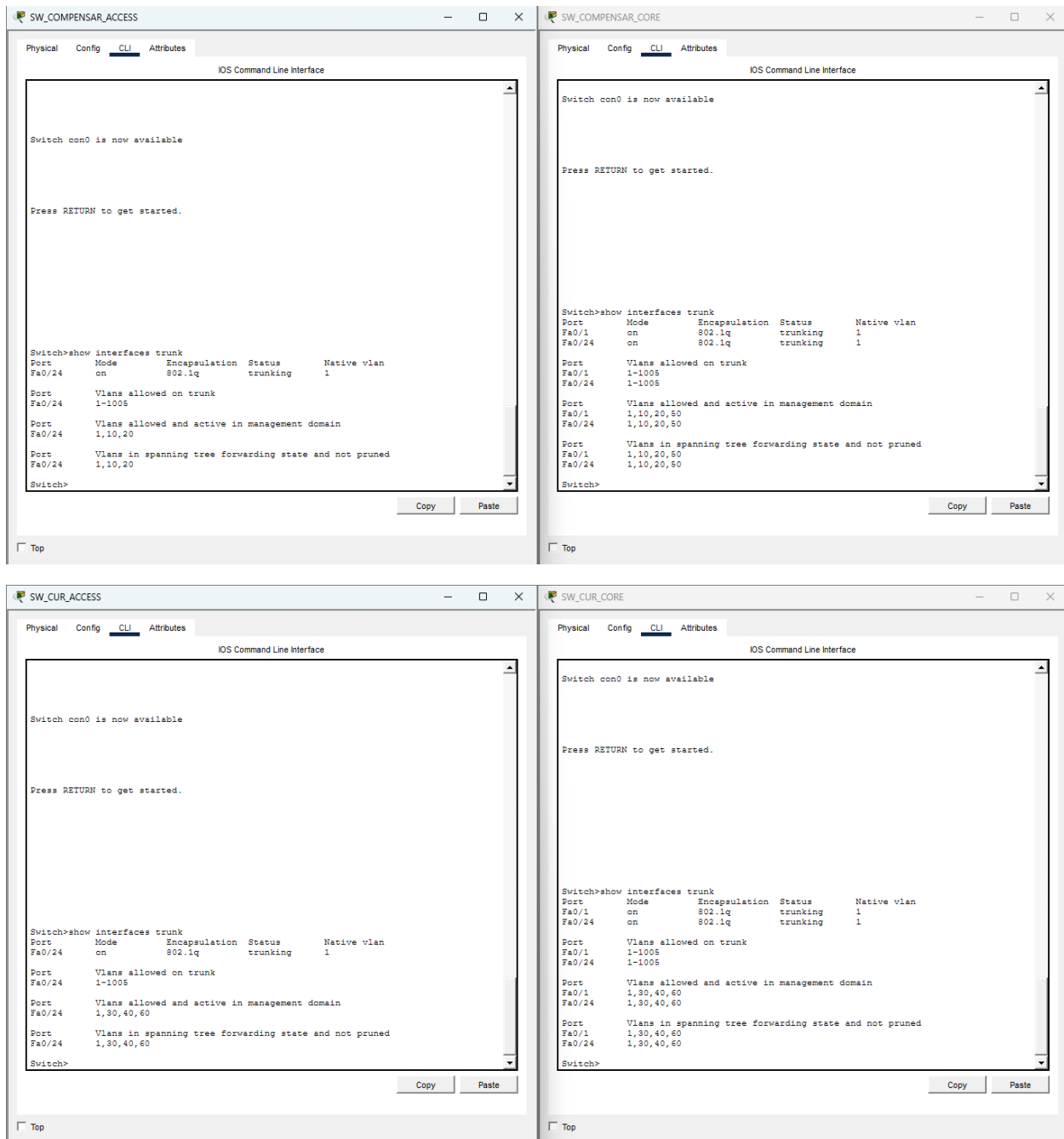
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                           Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Gig0/1, Gig0/2
30   DOCENTES                active
40   LABORATORIOS            active
80   CCTV                   active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch>

```

VALIDACIÓN DE TRONCALES



The image displays four screenshots of network switch CLI interfaces, arranged in a 2x2 grid. The top row shows the initial configuration and validation for switch SW_COMPENSAR_ACCESS, and the bottom row shows the same for switch SW_CUR_ACCESS.

Top Left: SW_COMPENSAR_ACCESS

Switch con0 is now available.

Press RETURN to get started.

Switch>show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/24	on	802.1q	trunking	1

Port Fa0/24 Vlans allowed on trunk 1-1005

Port Fa0/24 Vlans allowed and active in management domain 1,10,20

Port Fa0/24 Vlans in spanning tree forwarding state and not pruned 1,10,20

Switch>

Top Right: SW_COMPENSAR_CORE

Switch con0 is now available.

Press RETURN to get started.

Switch>show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1
Fa0/24	on	802.1q	trunking	1

Port Fa0/1 Vlans allowed on trunk 1-1005

Port Fa0/24 Vlans allowed on trunk 1-1005

Port Fa0/1 Vlans allowed and active in management domain 1,10,20,80

Port Fa0/24 Vlans allowed and active in management domain 1,10,20,80

Port Fa0/1 Vlans in spanning tree forwarding state and not pruned 1,10,20,80

Port Fa0/24 Vlans in spanning tree forwarding state and not pruned 1,10,20,80

Switch>

Bottom Left: SW_CUR_ACCESS

Switch con0 is now available.

Press RETURN to get started.

Switch>show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/24	on	802.1q	trunking	1

Port Fa0/24 Vlans allowed on trunk 1-1005

Port Fa0/24 Vlans allowed and active in management domain 1,30,40,60

Port Fa0/24 Vlans in spanning tree forwarding state and not pruned 1,30,40,60

Switch>

Bottom Right: SW_CUR_CORE

Switch con0 is now available.

Press RETURN to get started.

Switch>show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1
Fa0/24	on	802.1q	trunking	1

Port Fa0/1 Vlans allowed on trunk 1-1005

Port Fa0/24 Vlans allowed on trunk 1-1005

Port Fa0/1 Vlans allowed and active in management domain 1,30,40,60

Port Fa0/24 Vlans allowed and active in management domain 1,30,40,60

Port Fa0/1 Vlans in spanning tree forwarding state and not pruned 1,30,40,60

Port Fa0/24 Vlans in spanning tree forwarding state and not pruned 1,30,40,60

Switch>

VALIDACIÓN DE SUBINTERFACES

RTR_COMPENSAR_CORE

Physical Config **CLI** Attributes

IOS Command Line Interface

Router com0 is now available

Press RETURN to get started.

Router>show interfaces trunk

Router>

Router>show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0.50	192.168.50.1	YES	manual	up	up
GigabitEthernet0/1	10.10.10.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
FastEthernet0/0/0	unassigned	YES	unset	down	down
FastEthernet0/0/1	unassigned	YES	unset	down	down
FastEthernet0/0/2	unassigned	YES	unset	down	down
FastEthernet0/0/3	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down

Router>

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Top

RTR_CUR_CORE

Physical Config **CLI** Attributes

IOS Command Line Interface

Router com0 is now available

Press RETURN to get started.

Router>show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/0.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/0.60	192.168.60.1	YES	manual	up	up
GigabitEthernet0/1	10.10.10.2	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
FastEthernet0/0/0	unassigned	YES	unset	down	down
FastEthernet0/0/1	unassigned	YES	unset	down	down
FastEthernet0/0/2	unassigned	YES	unset	down	down
FastEthernet0/0/3	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down

Router>

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VALIDACIÓN DEL DHCP EN CADA EQUIPO

PC_ADMIN

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.10.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 192.168.50.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::240:8FF:FECA:A859

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC_ACADEMICA

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.20.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 192.168.50.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:63FF:FEED:E383

Default Gateway

DNS Server

802.1X

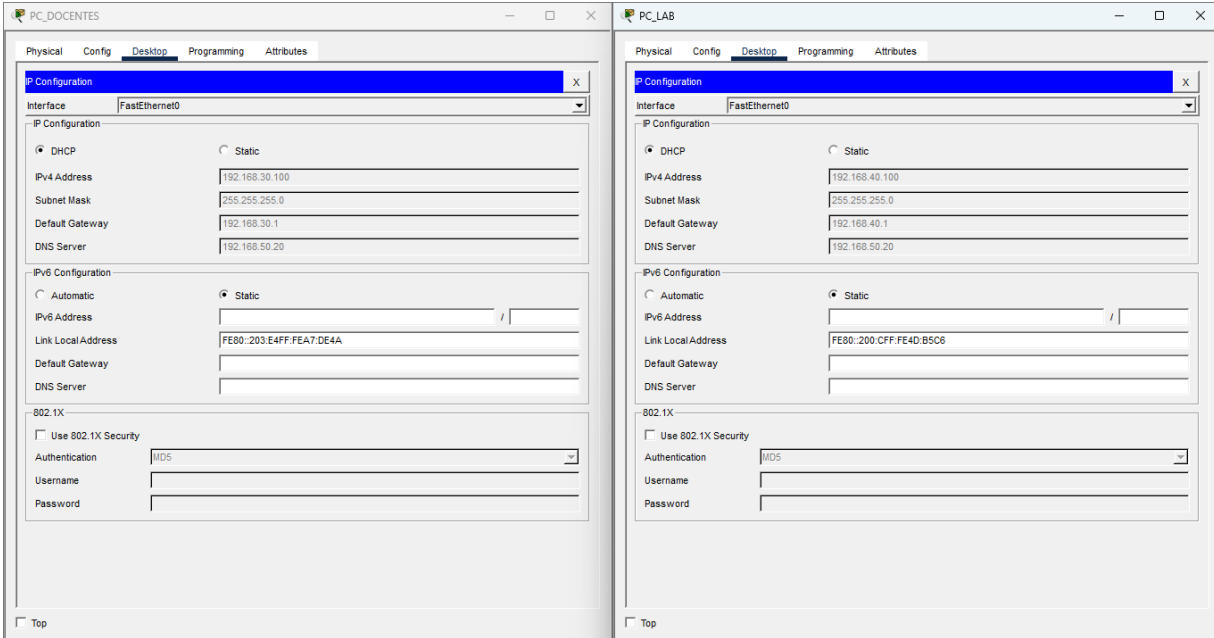
☐ Use 802.1X Security

Authentication MD5

Username

Password

Top



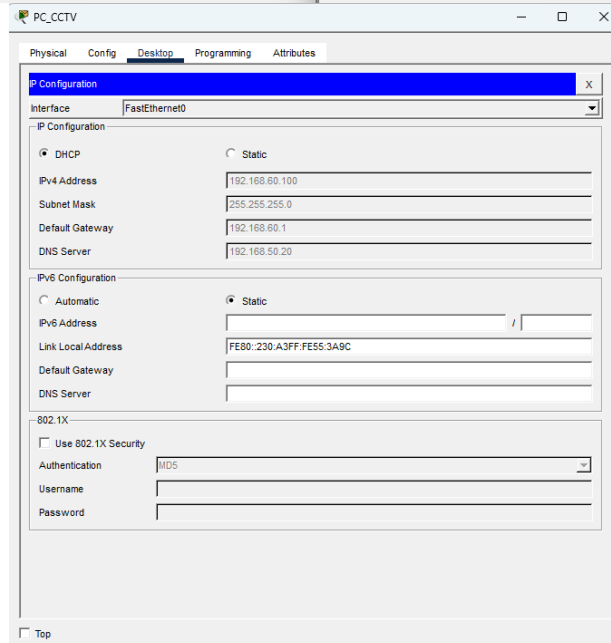
PC configuration interface showing IP Configuration for FastEthernet0. The interface is divided into IPv4 Configuration, IPv6 Configuration, and 802.1X Security sections.

PC_DOCENTES:

- Interface: FastEthernet0
- IPv4 Configuration: DHCP selected, IPv4 Address: 192.168.30.100, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.30.1, DNS Server: 192.168.50.20
- IPv6 Configuration: Static selected, IPv6 Address: /, Link Local Address: FE80::203:E4FF:FEA7:DE4A, Default Gateway: , DNS Server:
- 802.1X Security: Use 802.1X Security unchecked, Authentication: MD5, Username: , Password:

PC_LAB:

- Interface: FastEthernet0
- IPv4 Configuration: DHCP selected, IPv4 Address: 192.168.40.100, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.40.1, DNS Server: 192.168.50.20
- IPv6 Configuration: Static selected, IPv6 Address: /, Link Local Address: FE80::200:CFF:FE4D:B5C6, Default Gateway: , DNS Server:
- 802.1X Security: Use 802.1X Security unchecked, Authentication: MD5, Username: , Password:



PC configuration interface showing IP Configuration for FastEthernet0. The interface is divided into IPv4 Configuration, IPv6 Configuration, and 802.1X Security sections.

PC_CCTV:

- Interface: FastEthernet0
- IPv4 Configuration: DHCP selected, IPv4 Address: 192.168.60.100, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.60.1, DNS Server: 192.168.50.20
- IPv6 Configuration: Static selected, IPv6 Address: /, Link Local Address: FE80::230:A3FF:FE55:3A9C, Default Gateway: , DNS Server:
- 802.1X Security: Use 802.1X Security unchecked, Authentication: MD5, Username: , Password:

VALIDACIÓN DE CONECTIVIDAD LOCAL

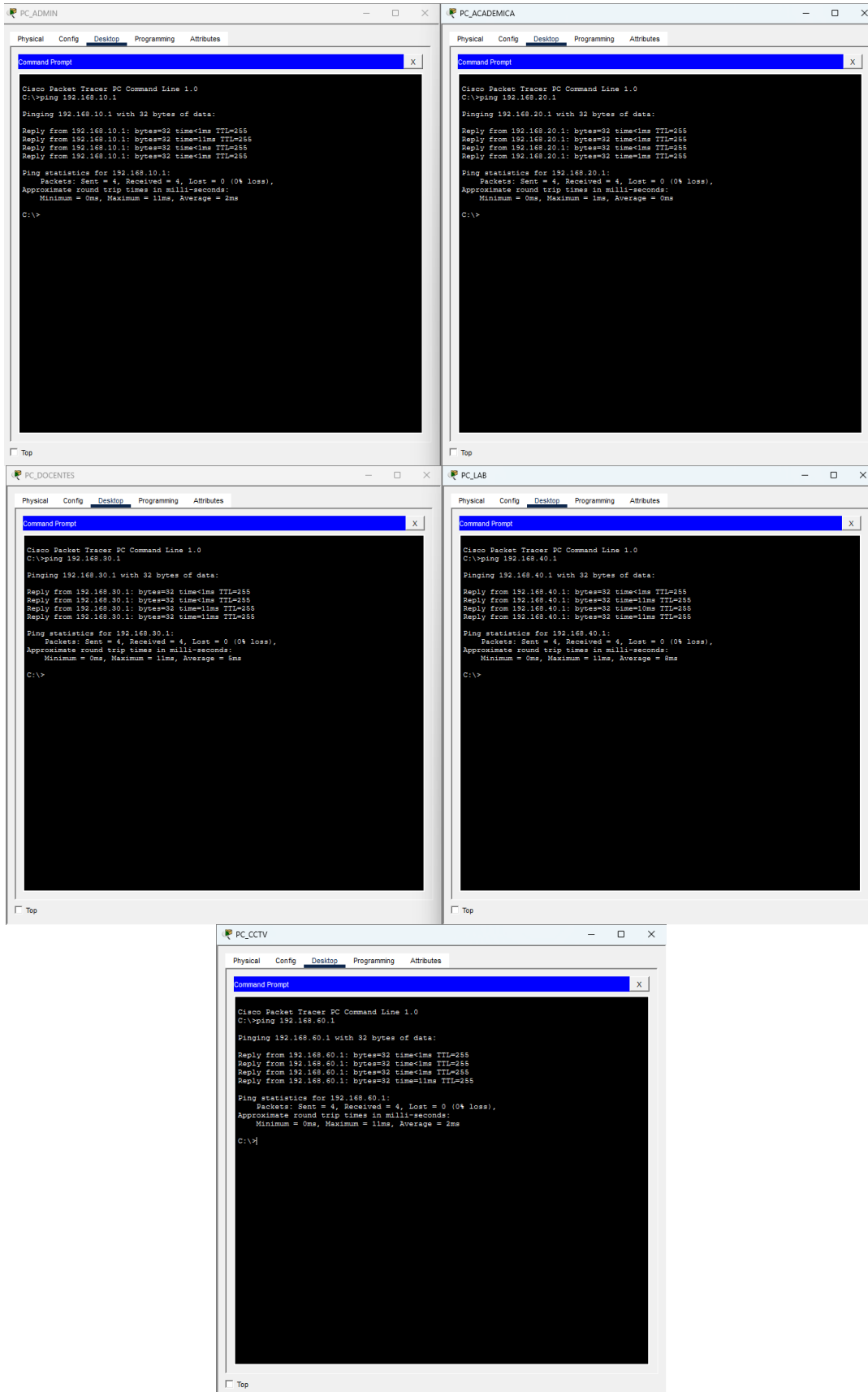
PC ADMIN
ping 192.168.10.1

PC LABORATORIOS
ping 192.168.40.1

PC ACADEMIA
ping 192.168.20.1

PC CCTV
ping 192.168.60.1

PC DOCENTES
ping 192.168.30.1



The image displays five Cisco Packet Tracer PC Command Line windows, each showing the output of a ping command. The windows are titled PC_ADMIN, PC_ACADEMICA, PC_DOCENTES, PC_LAB, and PC_CCTV. Each window shows the results of a ping command to a specific IP address, including the number of bytes, TTL, and round trip times.

PC_ADMIN

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 2ms
C:\>
```

PC_ACADEMICA

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

PC_DOCENTES

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 8ms
C:\>
```

PC_LAB

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 8ms
C:\>
```

PC_CCTV

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.60.1

Pinging 192.168.60.1 with 32 bytes of data:

Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.60.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 2ms
C:\>
```

VALIDACIÓN DEL OSPF

RTR_COMPENSAR_CORE

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
192.168.60.1    1     FULL/DR         00:00:36    10.10.10.2     GigabitEthernet0/1

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.10.10.0/30 is directly connected, GigabitEthernet0/1
L    10.10.10.1/32 is directly connected, GigabitEthernet0/1
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
C    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
O    192.168.30.0/24 [110/2] via 10.10.10.2, 01:08:57, GigabitEthernet0/1
O    192.168.40.0/24 [110/2] via 10.10.10.2, 01:08:57, GigabitEthernet0/1
C    192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L    192.168.50.1/32 is directly connected, GigabitEthernet0/0.50
O    192.168.60.0/24 [110/2] via 10.10.10.2, 01:08:57, GigabitEthernet0/1

Router>
Router>
Router>
Router>
```

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☐ Top

RTR_CUR_CORE

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
192.168.60.1    1     FULL/DR         00:00:34    10.10.10.1     GigabitEthernet0/1

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.10.10.0/30 is directly connected, GigabitEthernet0/1
L    10.10.10.1/32 is directly connected, GigabitEthernet0/1
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
O    192.168.10.0/24 [110/2] via 10.10.10.1, 01:09:06, GigabitEthernet0/1
O    192.168.20.0/24 [110/2] via 10.10.10.1, 01:09:06, GigabitEthernet0/1
C    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
C    192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L    192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
O    192.168.50.0/24 [110/2] via 10.10.10.1, 01:09:06, GigabitEthernet0/1
C    192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.60.0/24 is directly connected, GigabitEthernet0/0.60
L    192.168.60.1/32 is directly connected, GigabitEthernet0/0.60

Router>
Router>
Router>
Router>
```

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☐ Top

CONECTIVIDAD ENTRE UBICACIONES

PC_ADMIN

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>ping 192.168.30.100

Pinging 192.168.30.100 with 32 bytes of data:

Reply from 192.168.30.100: bytes=32 time<1ms TTL=126
Reply from 192.168.30.100: bytes=32 time=26ms TTL=126
Reply from 192.168.30.100: bytes=32 time=12ms TTL=126
Reply from 192.168.30.100: bytes=32 time=16ms TTL=126

Ping statistics for 192.168.30.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 26ms, Average = 16ms

C:\>
```

☐ Top

PC_DOCENTES

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=11ms TTL=255
Reply from 192.168.30.1: bytes=32 time=11ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 6ms

C:\>ping 192.168.10.100

Pinging 192.168.10.100 with 32 bytes of data:

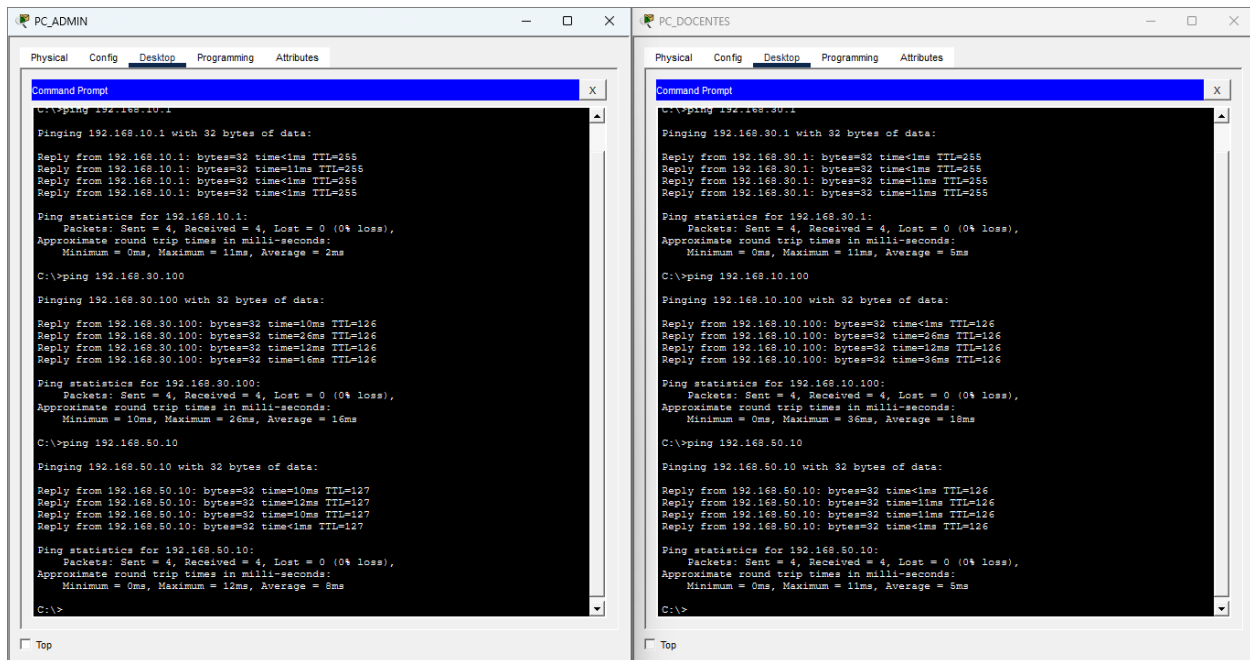
Reply from 192.168.10.100: bytes=32 time<1ms TTL=126
Reply from 192.168.10.100: bytes=32 time=26ms TTL=126
Reply from 192.168.10.100: bytes=32 time=12ms TTL=126
Reply from 192.168.10.100: bytes=32 time=36ms TTL=126

Ping statistics for 192.168.10.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 36ms, Average = 18ms

C:\>
```

☐ Top

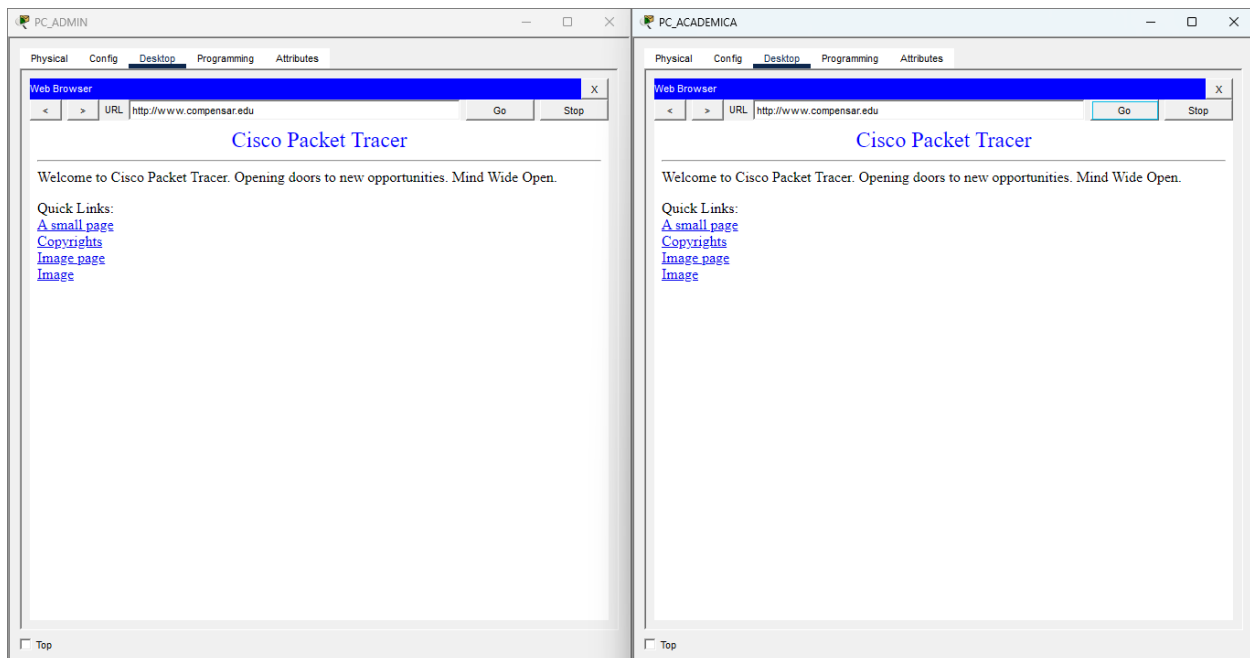
VALIDACIÓN DEL SERVIDOR DHCP



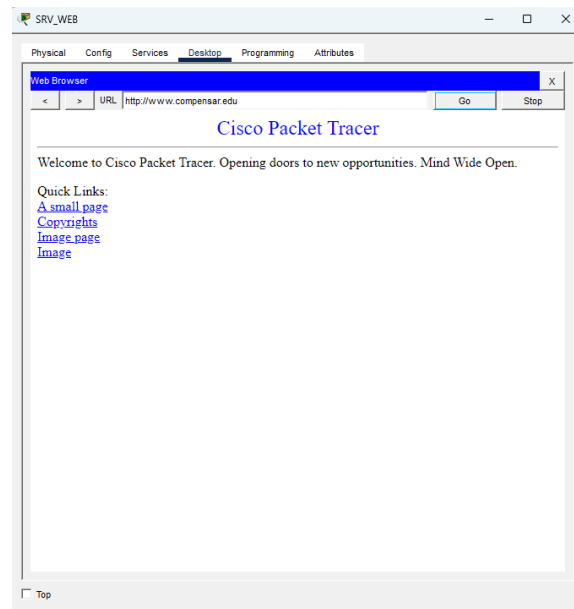
```
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms
C:\>ping 192.168.30.100
Pinging 192.168.30.100 with 32 bytes of data:
Reply from 192.168.30.100: bytes=32 time=10ms TTL=126
Reply from 192.168.30.100: bytes=32 time=46ms TTL=126
Reply from 192.168.30.100: bytes=32 time=12ms TTL=126
Reply from 192.168.30.100: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.30.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 26ms, Average = 16ms
C:\>ping 192.168.50.10
Pinging 192.168.50.10 with 32 bytes of data:
Reply from 192.168.50.10: bytes=32 time=10ms TTL=127
Reply from 192.168.50.10: bytes=32 time=12ms TTL=127
Reply from 192.168.50.10: bytes=32 time=10ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 8ms
C:\>
```

```
C:\>ping 192.168.30.1
Pinging 192.168.30.1 with 32 bytes of data:
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms
C:\>ping 192.168.10.100
Pinging 192.168.10.100 with 32 bytes of data:
Reply from 192.168.10.100: bytes=32 time<1ms TTL=126
Reply from 192.168.10.100: bytes=32 time=46ms TTL=126
Reply from 192.168.10.100: bytes=32 time=12ms TTL=126
Reply from 192.168.10.100: bytes=32 time=36ms TTL=126
Ping statistics for 192.168.10.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 36ms, Average = 18ms
C:\>ping 192.168.50.10
Pinging 192.168.50.10 with 32 bytes of data:
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time=1ms TTL=126
Reply from 192.168.50.10: bytes=32 time=1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 6ms
C:\>
```

VALIDACIÓN DEL DNS



VALIDACIÓN DEL SERVIDOR WEB



VALIDACIÓN DE SERVIDORES

DNS

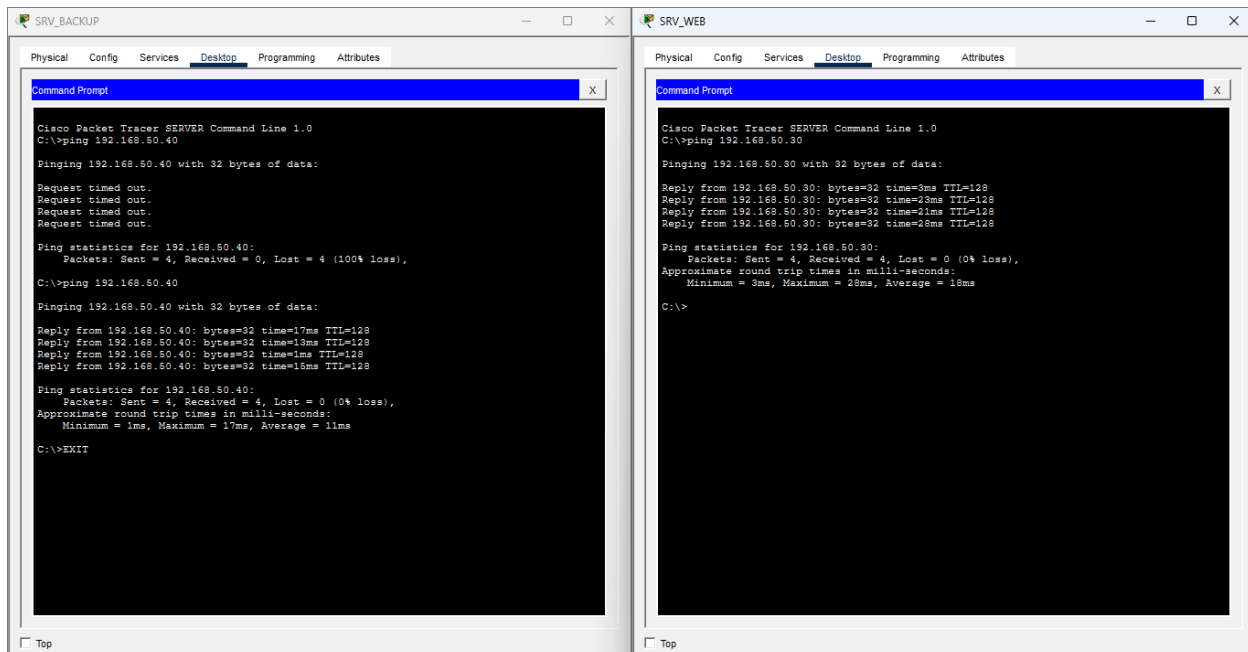
ping 192.168.50.20

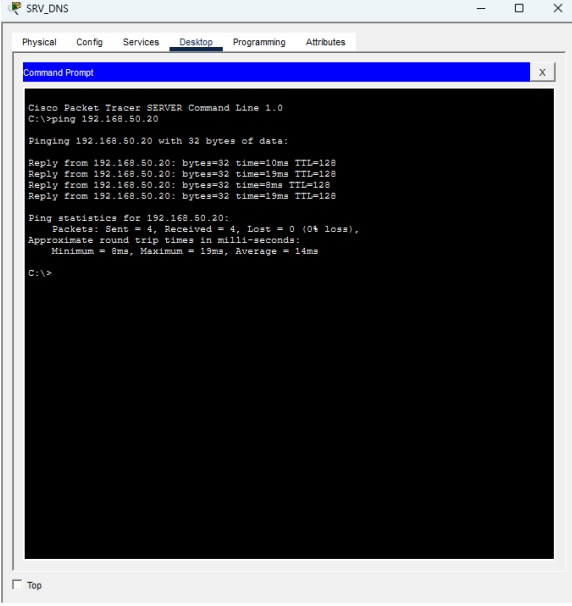
WEB

ping 192.168.50.30

BACKUP

ping 192.168.50.40





```
SRV_DNS
Physical Config Services Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.50.20

Pinging 192.168.50.20 with 32 bytes of data:

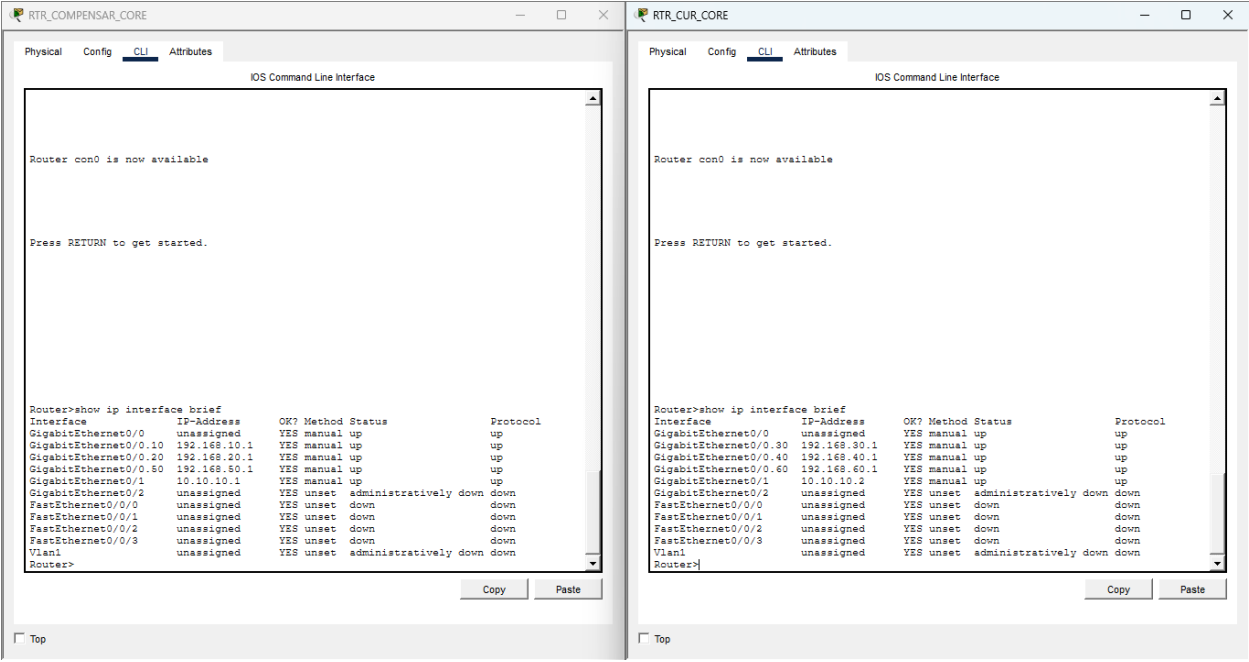
Reply from 192.168.50.20: bytes=32 time=10ms TTL=128
Reply from 192.168.50.20: bytes=32 time=10ms TTL=128
Reply from 192.168.50.20: bytes=32 time=8ms TTL=128
Reply from 192.168.50.20: bytes=32 time=19ms TTL=128

Ping statistics for 192.168.50.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 19ms, Average = 14ms

C:\>
```

PRUEBAS AVANZADAS DE SEGURIDAD, REDUNDANCIA Y DESEMPEÑO.

VALIDACION DE REDUNDANCIA



```
RTR_CUR_CORE
Physical Config CLI Attributes
IOS Command Line Interface

Router com0 is now available.

Press RETURN to get started.

Router>show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0  unassigned     YES manual up       up
GigabitEthernet0/0.10  192.168.10.1   YES manual up       up
GigabitEthernet0/0.20  192.168.20.1   YES manual up       up
GigabitEthernet0/0.50  192.168.50.1   YES manual up       up
GigabitEthernet0/1    10.10.10.1     YES manual up       up
GigabitEthernet0/2    unassigned     YES unset administratively down down
FastEthernet0/0/0     unassigned     YES unset down       down
FastEthernet0/0/1     unassigned     YES unset down       down
FastEthernet0/0/2     unassigned     YES unset down       down
FastEthernet0/0/3     unassigned     YES unset down       down
Vlan1             unassigned     YES unset administratively down down
Router>
```

VERIFICACIÓN DE RUTAS INSTALADAS

RTR_COMPENSAR_CORE

Physical Config CLI Attributes

IOS Command Line Interface

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.10.10.0/30 is directly connected, GigabitEthernet0/1
L 10.10.10.1/32 is directly connected, GigabitEthernet0/1
O 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
O 192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L 192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
O 192.168.30.0/24 [110/2] via 10.10.10.2, 02:23:11, GigabitEthernet0/1
L 192.168.40.0/24 [110/2] via 10.10.10.2, 02:23:11, GigabitEthernet0/1
O 192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L 192.168.50.1/32 is directly connected, GigabitEthernet0/0.50
O 192.168.60.0/24 [110/2] via 10.10.10.2, 02:23:11, GigabitEthernet0/1

Router>
Router>
Router>

Copy Paste

RTR_CUR_CORE

Physical Config CLI Attributes

IOS Command Line Interface

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.10.10.0/30 is directly connected, GigabitEthernet0/1
L 10.10.10.1/32 is directly connected, GigabitEthernet0/1
O 192.168.10.0/24 [110/2] via 10.10.10.1, 02:23:20, GigabitEthernet0/1
C 192.168.20.0/24 [110/2] via 10.10.10.1, 02:23:20, GigabitEthernet0/1
O 192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L 192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
O 192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L 192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
O 192.168.50.0/24 [110/2] via 10.10.10.1, 02:23:20, GigabitEthernet0/1
C 192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.60.0/24 is directly connected, GigabitEthernet0/0.60
L 192.168.60.1/32 is directly connected, GigabitEthernet0/0.60

Router>
Router>
Router>

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VALIDACIÓN DE METRICAS

LATENCIA

PC_ADMIN

Physical Config Desktop Programming Attributes

Command Prompt

C:\>ping 192.168.50.40

Pinging 192.168.50.40 with 32 bytes of data:

Reply from 192.168.50.40: bytes=32 time<1ms TTL=127
Reply from 192.168.50.40: bytes=32 time<1ms TTL=127
Reply from 192.168.50.40: bytes=32 time<1ms TTL=127
Reply from 192.168.50.40: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.40:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 12ms, Average = 6ms

C:\>ping 192.168.50.30

Pinging 192.168.50.30 with 32 bytes of data:

Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.30:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 15ms, Average = 9ms

C:\>ping 192.168.50.30

Pinging 192.168.50.30 with 32 bytes of data:

Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.30:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 24ms, Average = 12ms

C:\>

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PC_ACADEMICA

Physical Config Desktop Programming Attributes

Command Prompt

C:\>ping 192.168.50.40

Pinging 192.168.50.40 with 32 bytes of data:

Reply from 192.168.50.40: bytes=32 time<1ms TTL=127
Reply from 192.168.50.40: bytes=32 time<1ms TTL=127
Reply from 192.168.50.40: bytes=32 time<1ms TTL=127
Reply from 192.168.50.40: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.40:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 41ms, Average = 13ms

C:\>ping 192.168.50.30

Pinging 192.168.50.30 with 32 bytes of data:

Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.30:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>ping 192.168.50.30

Pinging 192.168.50.30 with 32 bytes of data:

Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127
Reply from 192.168.50.30: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.30:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>

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